Melih Barsbey

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SUMMARY

AI researcher focused on reliable and efficient machine learning under real-world constraints. Current research examines how robustness and safety of multimodal systems transform under resource-efficiency pressures, and how representation geometry mediates these effects. Research published in *Nature Medicine*, *NeurIPS*, *ICML*, *ICLR*, and *ICCV* among others. Further research highlights include mechanistic interpretability, learning theory and grokking, and using human-AI complementarity and noisy supervision for medical decision-making. Industry experience in research and deployment of ML systems in various domains, as team lead, individual contributor, and freelance consultant.

PROFESSIONAL EXPERIENCE

Imperial College London, London, United Kingdom

- Post-Doctoral Research Associate in Department of Computing. — Jul 2024 – Present
  - Research focus on synergies and trade-offs between robustness and compressibility in deep learning, with applications in multiple modalities.
  - Contributions to research on mechanistic interpretability, topological deep learning for drug discovery, grokking and learning theory.
  - Publications in ICLR (x3), ICCV, ICML, Medical Image Analysis.

DeepMind, London, United Kingdom

- Research scientist internship with Robust and Verified AI team. — Jun 2022 – Oct 2022
  - Research on developing a confidence-based deferral system that leads to robust human-AI hybrid performance with sparse medical data.
  - Research on a framework for training, performance evaluation, and expert reliability analysis under noisy annotations.

Freelance Machine Learning Consultant

- Freelance consultancy for multiple software companies on the research, development, and deployment of machine learning systems. — Jul 2019 – Jun 2024
  - Development and deployment of AutoML pipelines for demand forecasting.

P1M1 Software Consulting Inc., Istanbul, Turkey

- Senior Data Scientist, Team Lead — Sep 2017 – Jun 2019
  - Algorithm development, software design and implementation of the flagship data science product, conducting cash transaction forecasting in over 8000 ATMs.

EDUCATION

Boğaziçi University, Istanbul, Turkey

- Ph.D. in Computer Engineering — Sep 2016 – Sep 2023
  - Focus: Machine learning, Bayesian statistics
  - Advisors: Prof. A. Taylan Cemgil and Assoc. Prof. Arzucan Özgür
  - Cumulative GPA: 4.00 / 4.00
- B.A., M.A. in Psychology — Sep 2007 – Aug 2014
  - Focus: Embodied language processing, causal explanations
  - Thesis: Perceptual simulations beyond the sentence: the effect of narrative shifts on perceptual simulations

SELECTED PUBLICATIONS AND PREPRINTS

[19] **Barsbey, M.**, Stevinson, E., Prieto, L., & Birdal, T. Vulnerability to Jailbreak and Prompt Injection Attacks under Activation Quantization: A Geometric Account. *Under review*, 2026.

[18] **Barsbey, M.**, Ribeiro, A. H., Şimşekli, U., & Birdal, T. On the Interaction of Compressibility and Adversarial Robustness. *International Conference on Learning Representations (ICLR)*, 2026.

[17] Stevinson, E., Prieto, L., **Barsbey, M.**, & Birdal, T. Adversarial Vulnerability from Interference Between Features in Superposition. *International Conference on Machine Learning (ICML)*, 2026.

[16] Prieto, L., Stevinson, E., **Barsbey, M.**, Birdal, T., & Mediano, P. A. M. Correlations in the Data Lead to Semantically Rich Feature Geometry Under Superposition. *International Conference on Learning Representations (ICLR)*, 2026.

[15] **Barsbey, M.**, Prieto, L., Zafeiriou, S., & Birdal, T. Large Learning Rates Simultaneously Achieve Robustness to Spurious Correlations and Compressibility. *International Conference on Computer Vision (ICCV)*, 2025.

[14] Prieto, L., **Barsbey, M.**, Mediano, P. A. M., & Birdal, T. Grokking at the Edge of Numerical Stability. *International Conference on Learning Representations (ICLR)*, 2025.

[13] Stutz, D.[*], Cemgil, A.T.[*], Roy, A.G.[*], Matejovicova, T.[*], **Barsbey, M.[*]**, et al. Evaluating AI Systems under Uncertain Ground Truth: A Case Study in Dermatology. *Medical Image Analysis*, *103*, p.103556, 2025. [*]Equal contribution.

[12] Wan, Y.[*], **Barsbey, M.[*]**, Zaidi, A., & Şimşekli, U. Implicit Compressibility of Overparametrized Neural Networks Trained with Heavy-Tailed SGD. *International Conference on Machine Learning (ICML)*, 2024. [*]Equal contribution

[11] Dvijotham, K.[*], Winkens, J.[*], **Barsbey, M.[*]**, Ghaisas, S.[*], Pawlowski, N. et al. Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians (CoDoC). *Nature Medicine*, 29 (7), 1814-1820, 2023. [*]Equal contribution.

[10] **Barsbey, M.**, & Cemgil, A. T. Modeling Hierarchical Seasonality through Low-Rank Tensor Decompositions in Time Series Analysis. *IEEE Access*, 11, 85770-85784, 2023.

[9] **Barsbey, M.[*]**, Özçelik, R.[*], Bağ, A. et al. pydebiaseddta: A Computational Software for Training Robust Drug-Target Affinity Prediction Models. *Journal of Computational Biology*, 30 (11), 1240-1245, 2023. [*]Equal contribution.

[8] **Barsbey, M.** *Utilizing Nonnegative Tensor Factorization Methods for Inference, Model Selection, and Analysis in Supervised Learning*. PhD dissertation. Boğaziçi University, Istanbul, Turkey, Sep 2023.

[7] Raj, A., **Barsbey, M.**, Gürbüzbalaban, M., Zhu, L., & Şimşekli, U. Algorithmic Stability of Heavy-Tailed Stochastic Gradient Descent on Least Squares. *Conference on Algorithmic Learning Theory (ALT)*, 2023.

[6] **Barsbey, M.[*]**, Sefidgaran, M.[*], Erdogdu, M. A., Richard, G., & Şimşekli, U. Heavy Tails in SGD and Compressibility of Overparametrized Neural Networks. *Advances in Neural Information Processing Systems Conference (NeurIPS)*, 2021. [*]Equal contribution

[5] Yıldırım, S., Kurutmaz, M. B., **Barsbey, M.**, Şimşekli, U., & Cemgil, A. T. Bayesian Allocation Model: Marginal Likelihood-based Model Selection for Count Tensors. *IEEE Journal of Selected Topics in Signal Processing*, 15(3), 560-573, 2021.

[4] Kurutmaz, M. B.[*], **Barsbey, M.[*]**, Cemgil, A. T., Yıldırım, S., & Şimşekli, U. Bayesian Model Selection for Identifying Markov Equivalent Causal Graphs. Paper presented at *Symposium on Advances in Approximate Bayesian Inference (AABI)*, 2019. *Equal contribution

[3] **Barsbey, M.** *Perceptual Simulations beyond the Sentence: The Effect of Narrative Shifts on Perceptual Simulations*. Master's thesis. Boğaziçi University, Istanbul, Turkey, Aug 2014.

[2] **Barsbey, M.**, Sarıbay, S. A., Boduroğlu, A., & Zwaan, R. A. Perceptual simulations beyond the sentence: the effect of narrative shifts on perceptual simulations. Poster presented at the *Annual Conference on Embodied and Situated Language Processing*, Rotterdam, The Netherlands, 2014.

[1] **Barsbey, M.**, & Lombrozo, T., Linguistic differences in explanation requests and their effects on the evaluation of explanations: the case of English and Turkish. Poster presented at the annual meeting of the *Cognitive Science Society*, Berlin, Germany, 2013.

SCIENTIFIC SERVICE

Reviewer

- Neural Information Processing Systems (NeurIPS) 2019 – Present
- International Conference on Machine Learning (ICML) 2021 – Present
- International Conference on Learning Representations (ICLR) 2021 – Present
- Uncertainty in Artificial Intelligence (UAI) 2023 – Present
- Transactions on Machine Learning Research (TMLR) 2023 – Present
- Conference on Computer Vision and Pattern Recognition (CVPR) 2024 – Present

AWARDS & SCHOLARSHIPS

- Global Talent Visa - United Kingdom 2024 – 2029
- PhD Scholarship from The Scientific and Technological Research Council of Turkey (TÜBİTAK) 2016 – 2021